



Reducing kWh Energy and kW Demand Comfortably

Presentation Hosted by



March 16, 2023

About Lodging Technology

- * Inventor and pioneer in Infrared (body heat) occupancy sensors for HVAC energy management.
- * Unsurpassed leader in energy management solutions for the hospitality sector.
- * For more than 43 years, we have helped hotels, resorts, vacation timeshare facilities and other business sectors reduce utility costs and increase bottom line profits.
- * How do we do it? By bringing together leading edge technology and extensive experience in the lodging industry.
- * We offer comprehensive and customized solutions – not cookie-cutter responses – and personalized services second to none.
- * Although designed primarily for the lodging industry, our systems are equally applicable to other facilities – such as school classrooms, college dorms, offices, conference centers, and military lodging facilities.

Four Primary Factors Determine Room Energy Consumption & Costs

Listed in Order of Highest Impact on Utility Costs

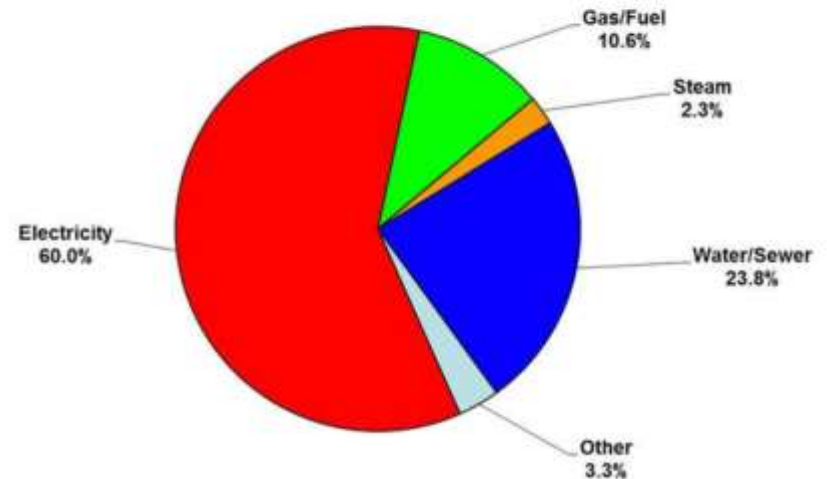
1. Climate (Primary Factor)
2. Local cost of energy per kWh, kW, and Gas Therm
3. Type of HVAC Unit
 - * PTAC / PTHP
 - * Split System
 - * Fan Coil Unit
4. Room Construction
 - * Exterior corridor
 - * High-rise full service interior corridor

(Note that Occupancy Rate is not listed and has Minimal Impact on most Property's Utility Costs)

Why should hoteliers and utilities be concerned about saving energy?

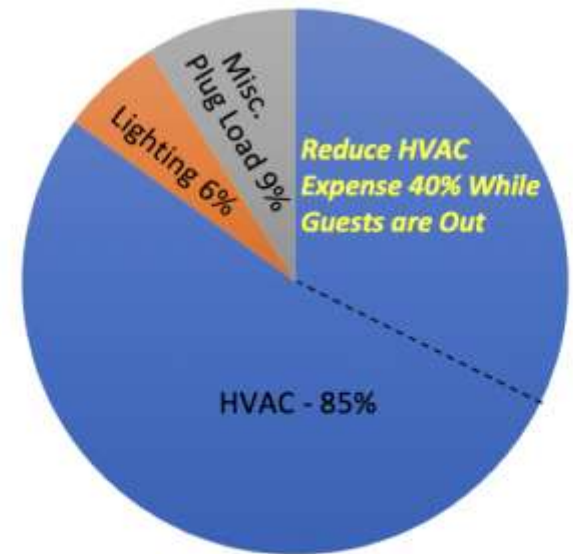
- * Cost of energy is **the** largest guestroom expense.
- * Total energy consumption is usually the second highest cost of business in the Hospitality industry.
- * Energy cost is typically 8% - 15% of a hotel's total operating costs.
- * Utilities must spend money to ensure enough generating, sub-station, wiring capacity to meet increasing demand.

2019
U.S. Lodging Industry
Mix of Utility Expenses -



Guestroom Energy Consumption...

- * HVAC (Heating, Ventilating, Air Conditioning)
 - * **Largest consumer of energy in the guestroom**
 - * Consumes 5-10 times more energy than lighting
 - * Costs \$18-\$40 per room each month depending on:
 - * Local climate
 - * Type of HVAC unit
 - * Room construction
 - * Utility rate



Guestroom Energy Use Profile

Reducing Energy Consumption:

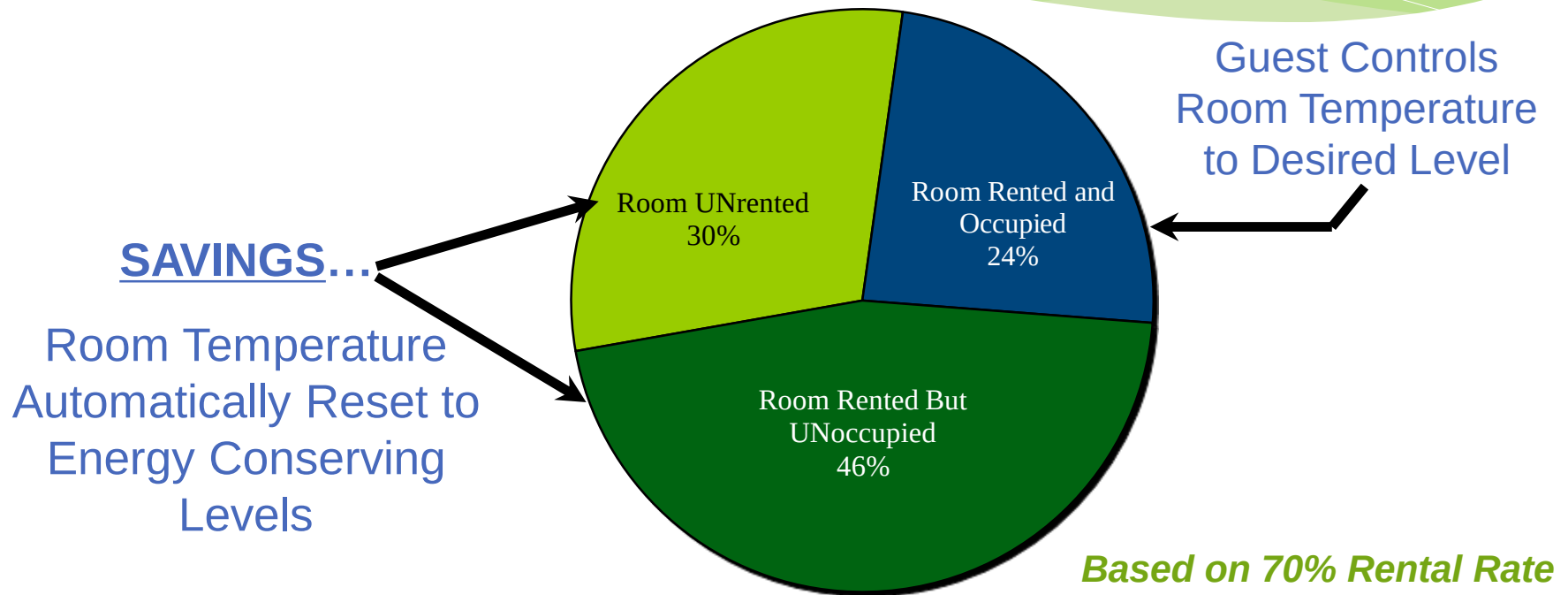
- * Hotels –
 - * Reduces operating costs
 - * Extends useful life of FF&E assets
 - * Adds profit to the bottom line
 - * Provides 50% - 60% Return on Investment (ROI) for owner
- * Duke Energy –
 - * Reduces kWh energy requirement
 - * Reduces kW Demand load on the Grid
 - * Reduces the need for larger sub-stations and new generating facilities
 - * Forestalls new government taxes and conservation mandates

How can guestroom Energy and Demand be significantly reduced?

- * ***Guestrooms are Empty . . .***

- * Unrented rooms are empty 100% of the time, except for staff
- * Rented rooms are empty 60% - 65% of a 24 hour day
- * HVAC, lights and TV continue to operate at guest-selected levels
- * Automatically reducing consumption when rooms are empty can save 35% - 45% in room utility costs
- * Savings can range from 1500 kWh – 2500 kWh or more per room each year
- * Reduce guestroom kW Demand 40% or more

Potential for Significant Savings



Let us show you how to reduce room kWh
Energy and kW Demand **35% to 45% ...**



with

 **GEMLink®**
WIRELESS
Energy Management

Cost Effective Guestroom HVAC Controls

- * **Properly Designed Occupancy Based Energy Management:**
 - * Determines physical presence of guests in rooms by detecting infrared body heat ; not simple motion
 - * Provides guests complete control of room environment while in the room
 - * Takes control when guests leave the room and resets temperature to energy conserving levels
 - * Reduces wasteful HVAC kWh consumption 35%-45%
 - * Reduces kW Demand

GEM Link® Wireless is Unique Among All Occupancy Sensor-Based Energy Management Systems

- GEM Link® is not a thermostat and does not connect to the thermostat;
- GEM Link® Wireless is “pure” energy management for ANY HVAC unit of any voltage with any type thermostat, including PTAC, PTHP, Split Systems, ductless Mini-Splits, 26” Window units, Fan Coil Units, or Electric Heaters;
- GEM Link® can also turn off selected Lighting in conference rooms, breakfast areas, classrooms, warehouses, as well as appliances such as the Two (2) Burner cooktop typically found in hotel suites when physically unoccupied to help prevent fire or sprinkler damage.
- GEM Link® does not require constant monitoring and adjustment.
- GEM Link® runs automatically in the background behind any thermostat and is not networked outside of the property, thus preventing malicious hackers from remotely accessing a property’s internal network or turning off HVAC units on guests.

Cost Effective Guestroom HVAC Controls

- * **EASY TO INSTALL AND PROGRAM**

- * Completely wireless and easy to install in 35 – 40 minutes without a computer or internet connection; superior battery life of 3 – 5 years; uses proprietary, modified 802.15.4 wireless protocol.

- * **VERSATILE**

- * Connects to **any** HVAC system, of **any** voltage, of **any** age, including 42” PTAC’s, Split Systems, 26” Window units, high mount Mini-Splits, Fan Coil Units, Electric Heat



Fan Coil Unit



PTAC / PTHP



26” Window Unit



Mini-Split



Split System



Electric Heat

Any HVAC of any age of any voltage with any Thermostat

(Utility Rebates available in certain areas.)

- * Designed and manufactured in the **United States** in Internationally Certified ISO 9001:2015 facilities

- *  Vietnam Era Veteran Owned Business



Applications

Applications Include Hotels, Schools, Condos, Offices, Warehouses, Military Lodging . . . and Anywhere there is an HVAC Unit of Any Kind in an Area that is Unoccupied for Periods of Time



Hotels / Motels



Guestrooms



Schools / Colleges



Dorm Rooms



Vacation Condos



Military Lodging

Home of National Security Agency,
U. S. Cyber Command



Component Description

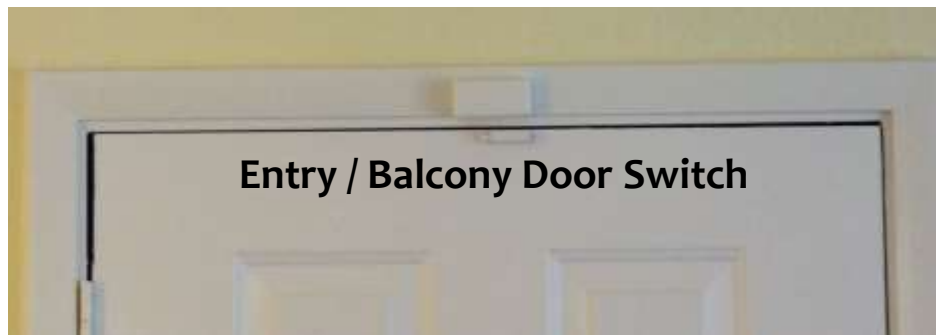
- * GEM Link® Wireless consists of **three basic components**:



Transceiver Control Module

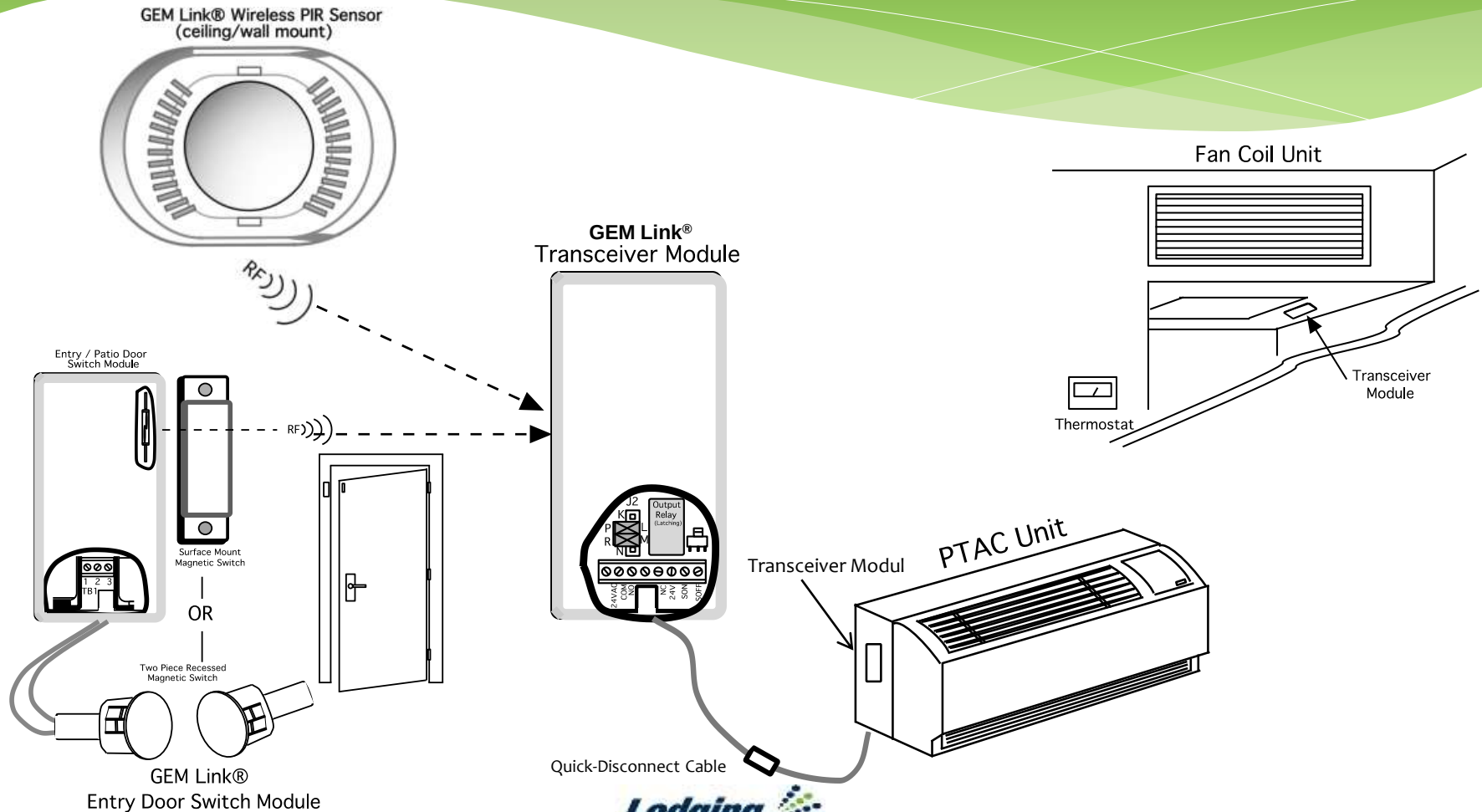


Passive Infrared (PIR) Sensor



Entry / Balcony Door Switch

Typical Connection of GEM Link® Wireless Components



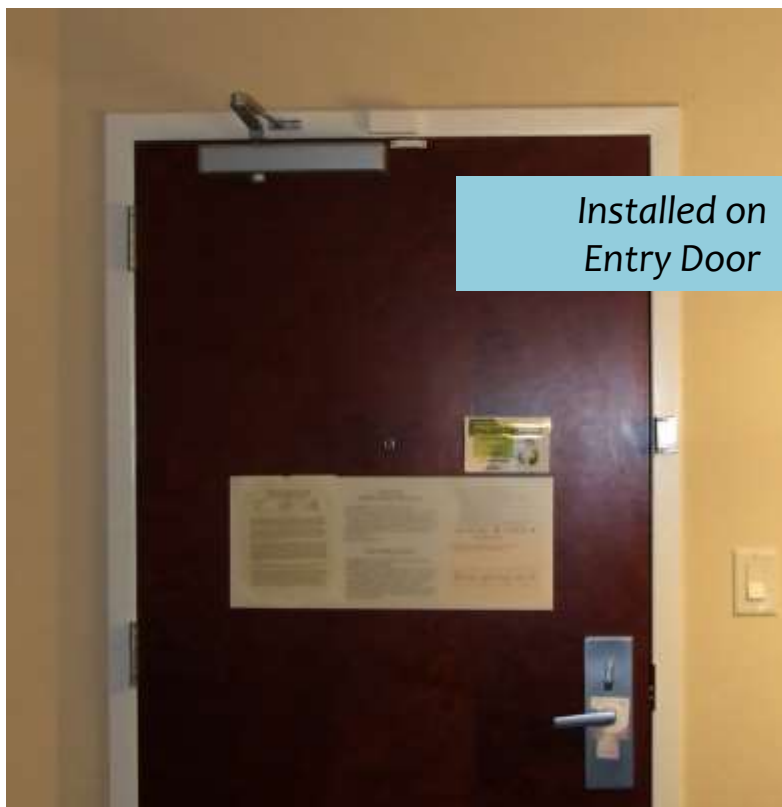
GEM Link® Actual Installation

Passive Infrared (PIR) Sensor Module



GEM Link[®] Actual Installation

Entry & Balcony Door Switch Module



*Installed on
Entry Door*



*Installed on Sliding
Balcony/Patio Door*

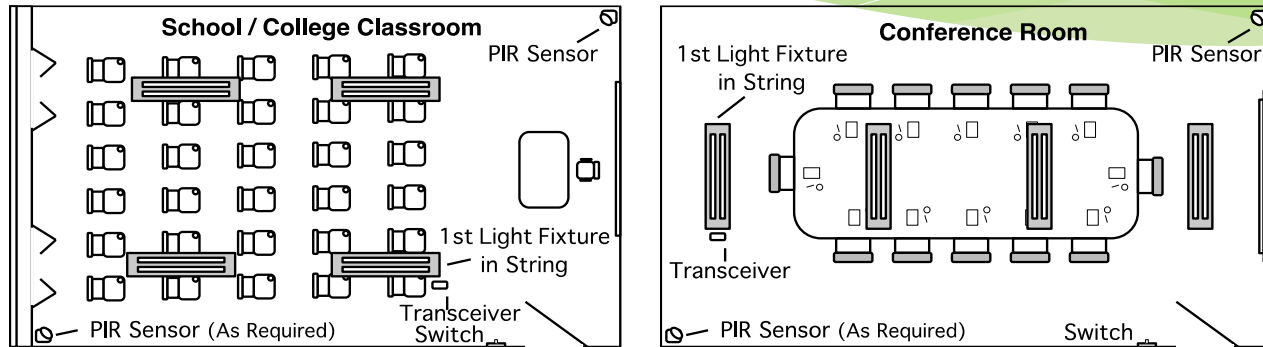
GEM Link[®] Actual Installation

Transceiver Control Module



GEM Link[®] - Other Applications

HVAC and Lighting Control in Classroom / Conference Room (Two Transceivers Sharing the Same PIR Sensors and Door Switches)



Two-Burner Cooktop



26" PTAC – No Wall Thermostat



Water Heater



Programmer / Maintenance Module ("PMM")

Hand-held device used to program system parameters during installation and monitor or change programmed features and perform maintenance tasks from outside guestroom without disturbing guests.

- * Wireless – no physical connection to GEM Link® system required
- * Sealed membrane keypad to resist damage from spills and dirt
- * Easy to understand – no special training or complicated procedures required



Room Status Scanner™ (“RSS”)

An Optional Feature of GEM Link® Wireless

Allows authorized hotel personnel to determine from the hallway whether a person is actually present inside a guestroom.

- * **STOPS** needlessly disturbing guests
- * Eliminates unwanted and embarrassing intrusion upon the guest while asleep, dressing, or in the shower
- * Improves staff efficiency – saves time & money
- * **Automatically tests GEM Link® operation, each day, ensuring maximum savings**



If someone is in the room, the “**OCCUPIED**” indicator on the RSS will illuminate **RED**.

If no one is in the room, the “**VACANT**” indicator on the RSS will illuminate **GREEN**.

Actual Savings



- * The following slides show typical savings our customers enjoy after installing GEM Link® Wireless from Lodging Technology

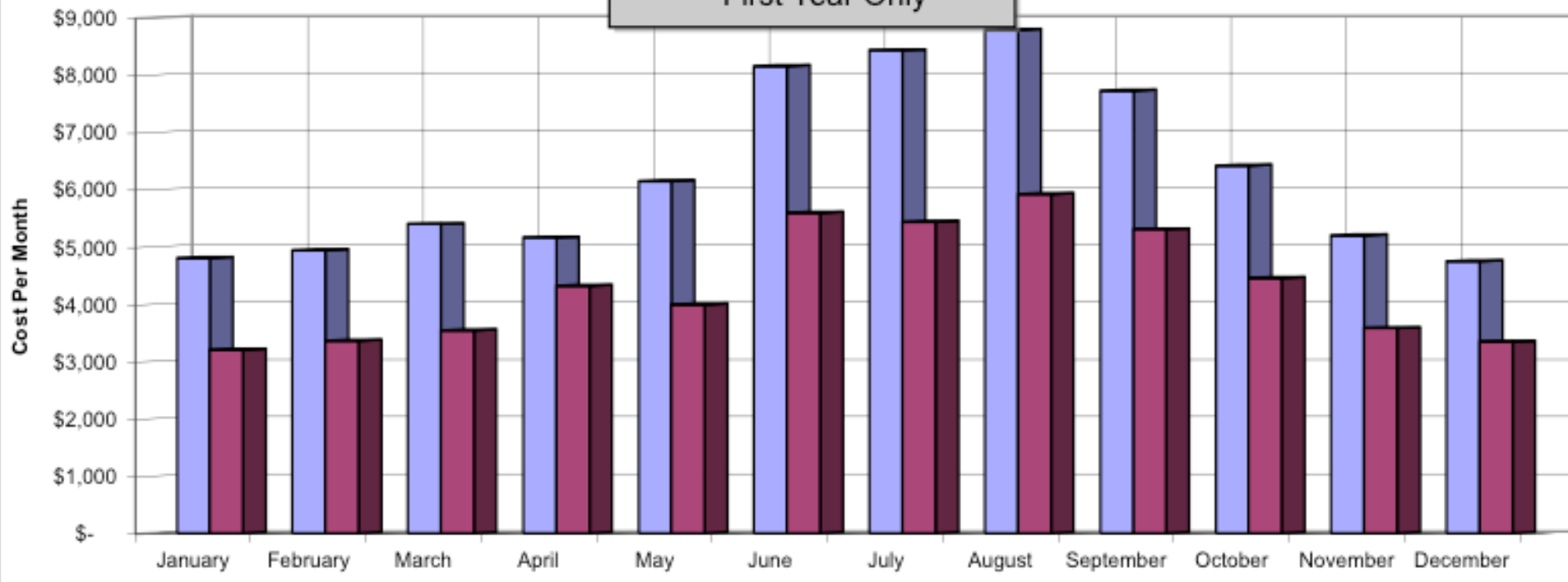
GEM Link® Wireless “Pure” Energy Management Saves at Least 20% MORE than Thermostat-Based Systems

Utility Bill Dollar Savings

Hampton Inn – 88 Rooms (Interior Corridor)

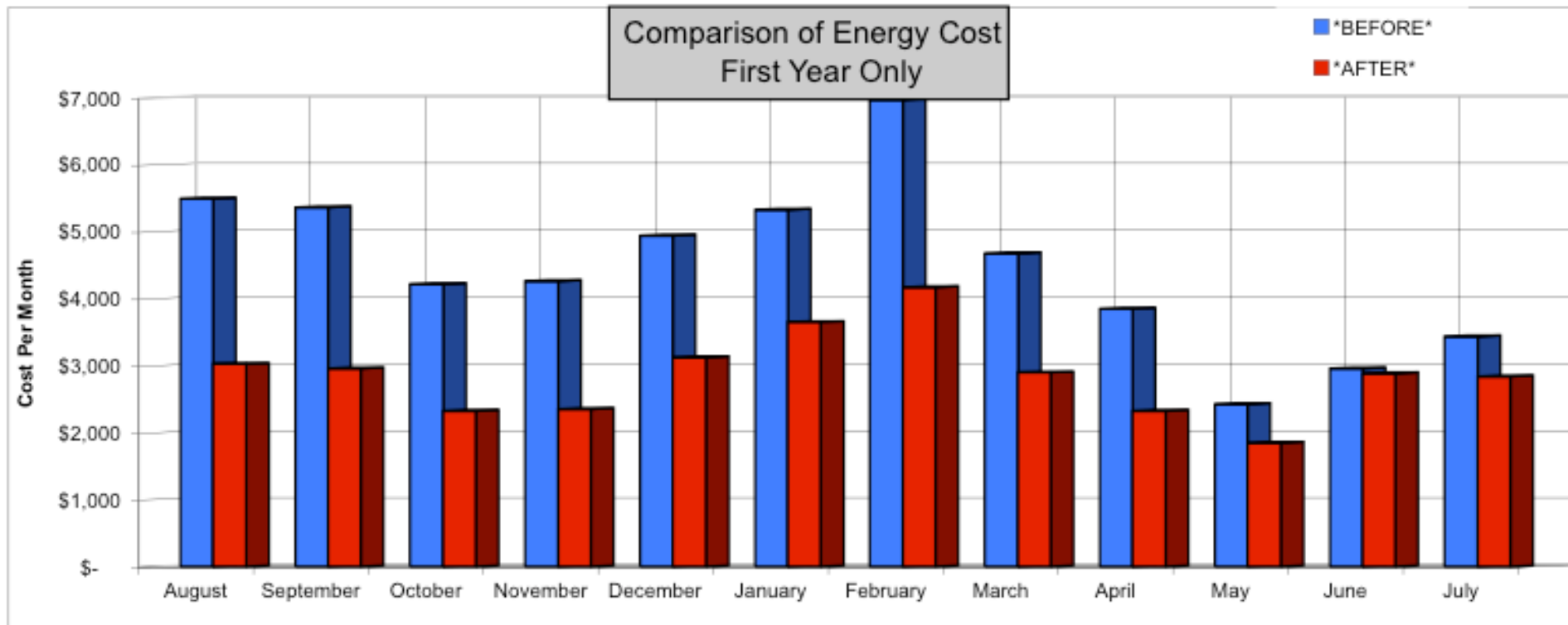
Comparison of Energy Cost
First Year Only

BEFORE
AFTER



Utility Bill Dollar Savings

Best Western – 73 Rooms (Exterior Corridor)



Hours of Operation Savings

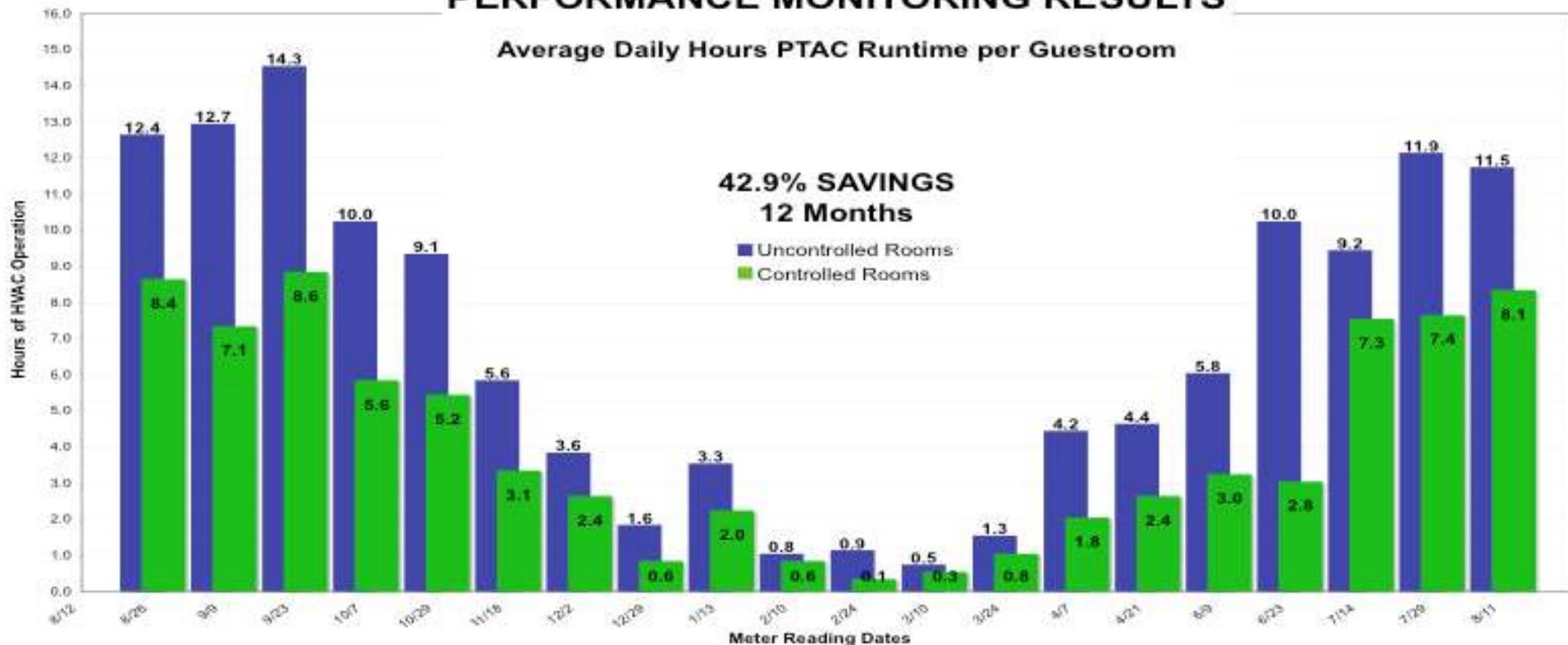
A Florida Resort Property

PERFORMANCE MONITORING RESULTS

Average Daily Hours PTAC Runtime per Guestroom

**42.9% SAVINGS
12 Months**

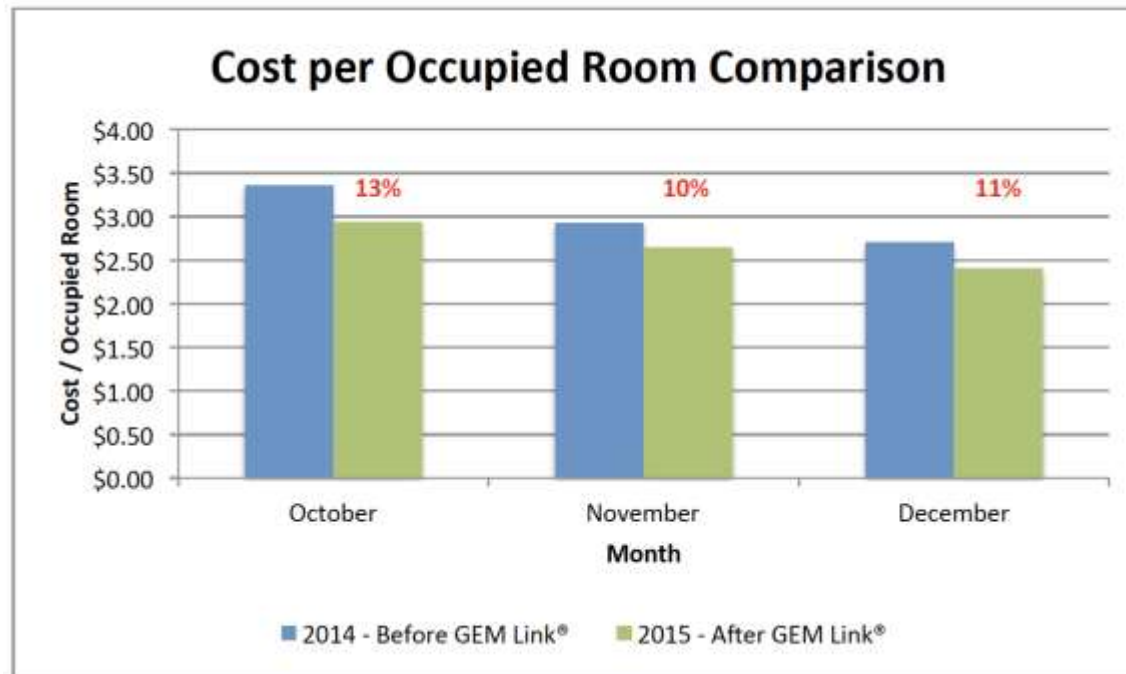
■ Uncontrolled Rooms
■ Controlled Rooms



Any Way You Measure It . . .

GEM Link® Wireless Reduces kWh and kW Demand

Cost per Occupied Room-	October	November	December
2014 (Before GEM Link®)	\$3.36	\$2.93	\$2.71
2015 (After GEM Link®)	\$2.94.	\$2.65	\$2.41
CPOR Savings- Total Property	13%	10%	11%



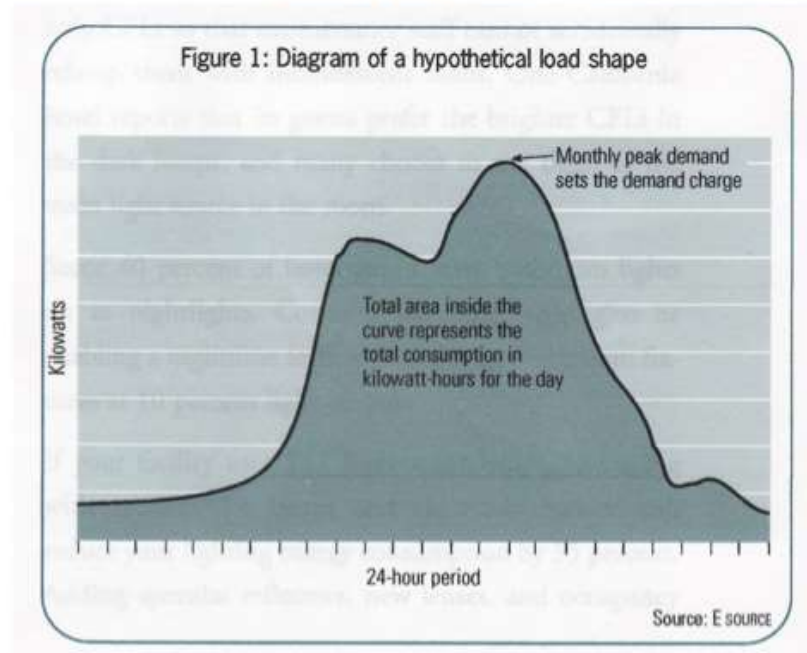
Typical Hotel Load Curve

Typical Hotel Load

“Most utilities charge commercial buildings for their natural gas based on the amount of energy delivered, but electricity can be charged based on consumption and demand (Figure 1).”

Direct

~ “Managing Energy Costs in Full-Service Hotels,” E Source Customer



Guestroom Occupancy Patterns

by Time of Day

University of California Davis CLTC Study 2011
(California Lighting Technology Center)

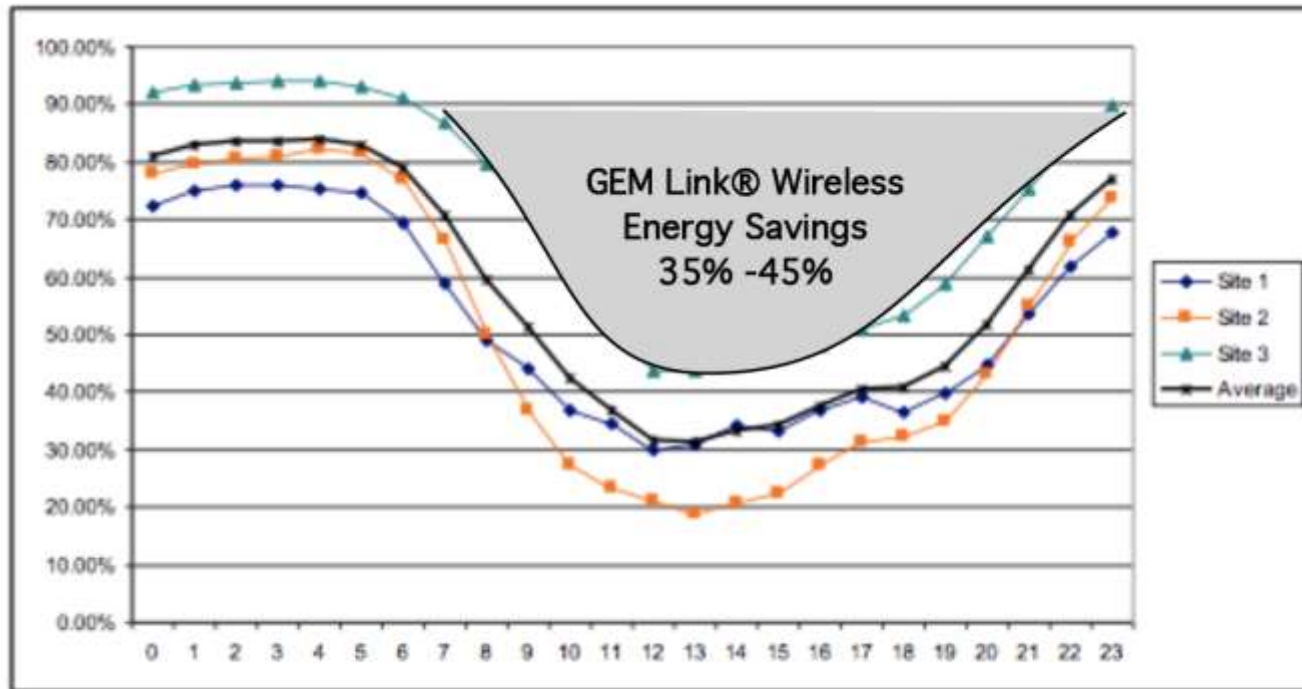


Figure 3: Occupancy Patterns of Hotel Guest Rooms from CLTC Field Study

kW Demand Savings

Massachusetts Electric

A **National Grid** Company



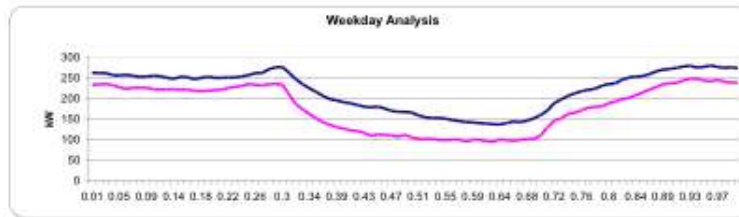
EPO Comparative Load Data Analysis

Hampton Inn

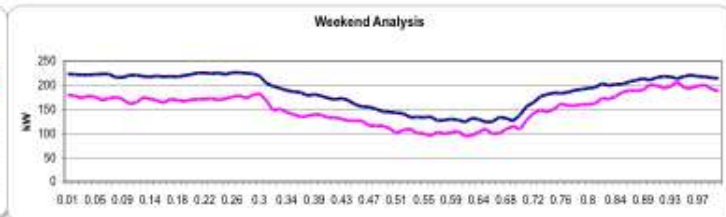
Jan-2004
Feb-2004

Before
After

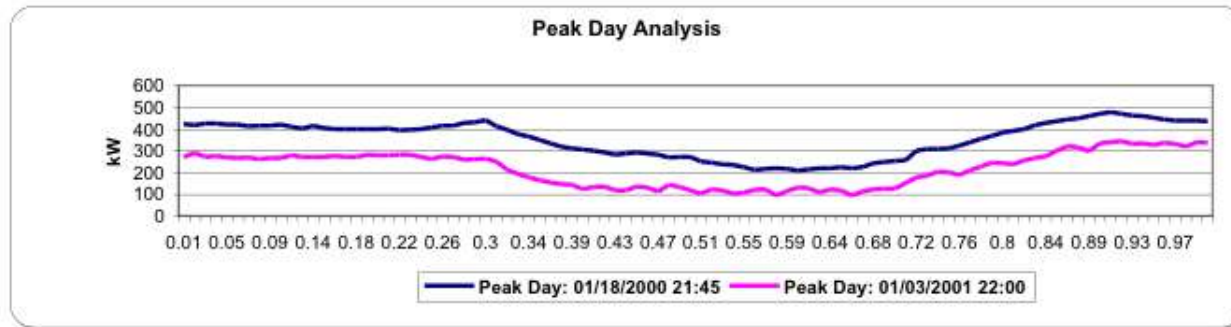
GEM System®



Increase/(Decrease) Cost Per Day (\$93)



Increase/(Decrease) Cost Per Day (\$78)



Increase/(Decrease) Peak Day (\$297)

Total Estimate Annual	
Increase/(Decrease) in Dollars	(\$32,394)
Annual % Increase/Decrease	-19%
Projected Annual Electric Bill	\$135,220

kW Demand Utility Bill Savings

- Northeast Beach Resort Property
 - kWh and kW Comparison June 2019 before GEM Link® and June 2021 after GEM Link®
 - Five Buildings with individual utility meters
 - Increased Occupancy June 2021, but reduced kWh, kW Demand, and Expense over 2019

17-Jun-19 NO CONTROLS						
	<u>DEMAND</u>					<u>Cost /</u>
<u>ACCOUNT</u>	<u>KWH</u>	<u>KWH</u>	<u>\$ / KWH</u>	<u>Cost</u>	<u># Days</u>	<u>Day</u>
4190023	1309	17.4	\$ 0.10	\$462.55	31	\$14.92
1810013	11240	40	\$ 0.10	\$3,007.03	31	\$97.00
1800014	11280	34	\$ 0.10	\$3,255.25	31	\$105.01
4170025	1093		\$ 0.10	\$331.63	30	\$11.05
4180024	1840	29.6	\$ 0.10	\$744.09	31	\$24.00
	26762			\$7,800.55		

16-Jun-21 GEM Link CONTROLS						
	<u>DEMAND</u>					
<u>ACCOUNT</u>	<u>KWH</u>	<u>KWH</u>	<u>\$ / KWH</u>	<u>Cost</u>	<u># Days</u>	<u>Cost / Day</u>
4190023	983	12	\$ 0.10	\$222.62	31	\$7.18
1810013	7640	34	\$ 0.10	\$1,559.45	31	\$50.30
1800014	11000	38	\$ 0.10	\$2,174.12	31	\$70.13
4170025	1104		\$ 0.10	\$237.21	28	\$8.47
4180024	2040	10	\$ 0.10	\$433.22	31	\$13.97
	22,767			\$4,626.62		

GEM Link® SAVINGS BY CATEGORY

	<u>DEMAND</u>		
<u>ACCOUNT</u>	<u>KWH</u>	<u>KWH</u>	<u>Cost</u>
4190023	24.90%	31.00%	51.90%
1810013	32.00%	15%	48.10%
1800014	2.50%	-11.80%	33.20%
4170025	-1.00%		28.50%
4180024	-10.80%	66.26%	41.80%
		40.70%	

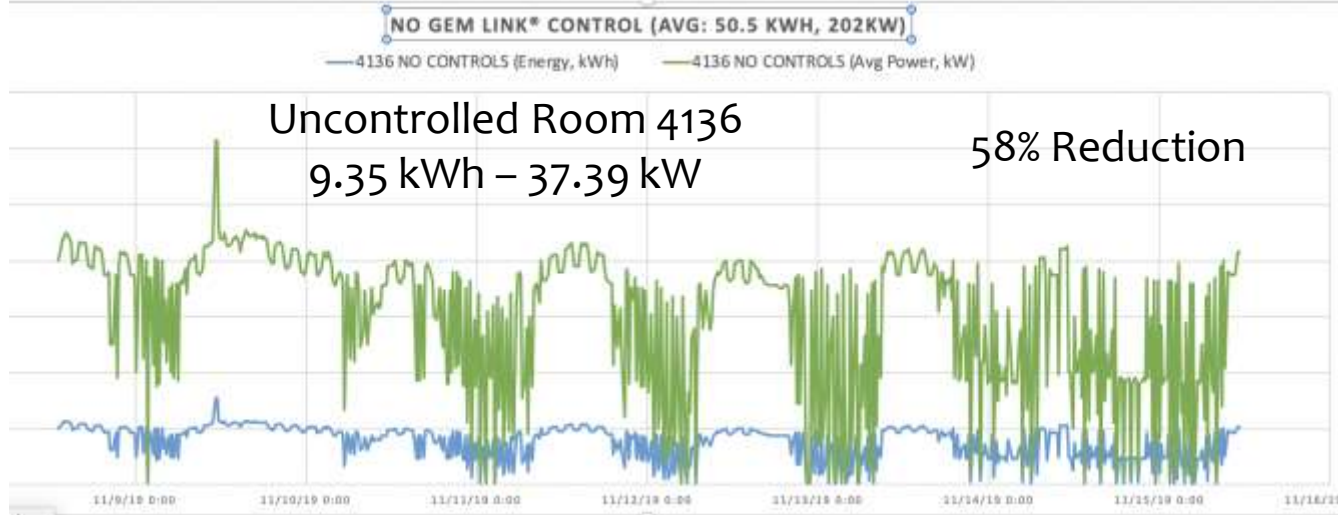
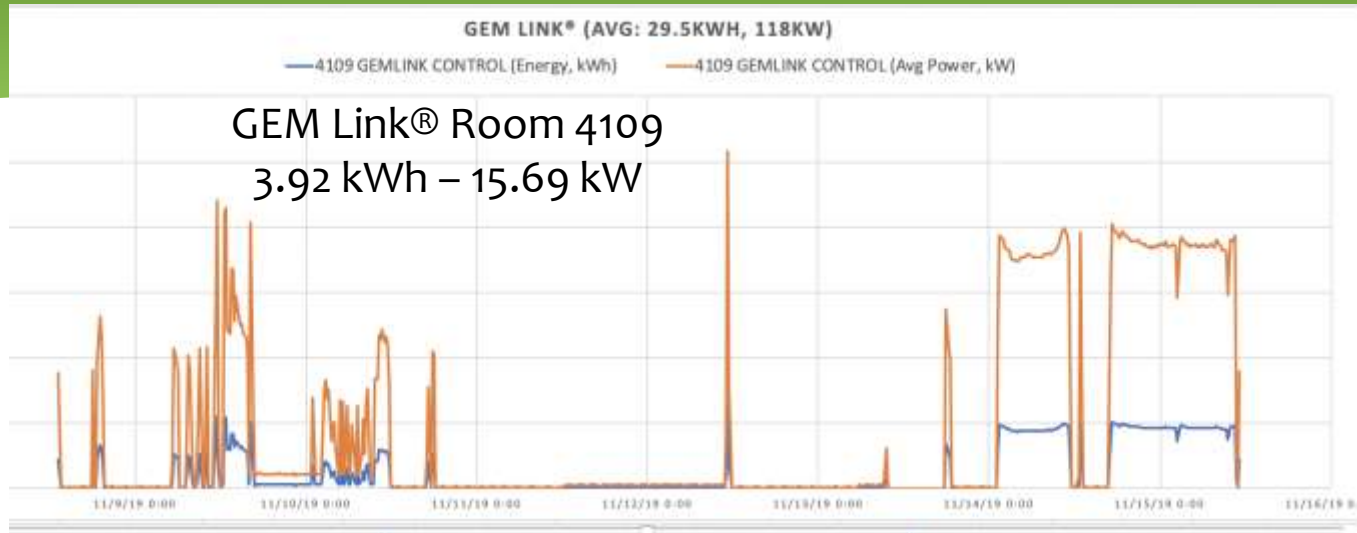
Note that kW Demand is generally lower so that the total invoice Cost is lower even though kWh may be equal or higher on the June 2021 invoices.

In Summary:

1. Total GEM Link® Wireless \$ Savings is \$3173.93 for June 2021 compared to June 2019;
2. Utility Cost per Day is lower for each invoice; and,
3. Average Savings for all invoices combined is 40.7% for the month.

GEM Link® kWh & kW vs Uncontrolled

Week of November 9, 2019



The Grove Resort & Spa Winter Garden, FL

HOBO Data Logger Record

kWh Energy Savings

